Unit 1 Kinematics

Textbook chapters 1,2,3

Next formative quiz will cover:
WB#2 position distance displacement speed velocity DT graphs without curves.pdf

Position, displacement, distance [Textbook 1.1]

Note & Practice: WB#2 pg1-5 position distance displacement.pdf

1 intro to vector motion lesson notesFILLED.pdf

✅ I know the difference between a VECTOR and a SCALAR (Scalars and Vectors)

✅ I know the difference between POSITION, DISPLACEMENT, and DISTANCE (Distance versus Displacement)

How to Solve a Vector Motion Problem:

Read question carefully and sketch a *line diagram* (draw an arrow to represent the motion interval)

Label the givens, unknowns (anything important that helps you understand the motion story); include and use a *direction convention*

Choose the formula/equation, rearrange if needed, substitute values (using your direction convention), solve and interpret your answer

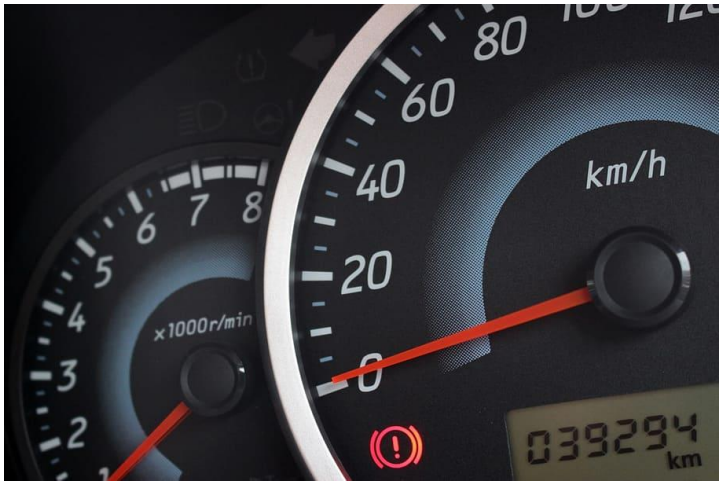
Homework problems WB pg4-5: 1b distance, displacement and position v2.pdf

Answers: 1d ANSWERS - distance, position, displacement 2024.pdf

Walk the Metronome Activity (video links below) 1.1 Speed note filled.pdf

Average Speed vs Instantaneous Speed vs Constant Speed (see middle section of graph):

Average speed = total distance travelled ➗ total time taken



Instantaneous speed:
e.g. what you read on a speedometer; the speed at an instant in time

When the speed is **constant**, the average speed will equal the instantaneous speed for any instant in time.

Average Speed vs Average Velocity [1.2]

Video: Speed and Velocity

WB 7-8 Speed vs Velocity.pdf 2a Ave speed vs Ave velocity note & answers.pdf

To put WB pg 8 answers here later…

WHITEBOARD ACTIVITY

Sketch the shape of a position-time graph of a moving object such that its average speed IS NOT the same as its average velocity.

Sketch the shape of a position-time graph of a moving object such that its average speed IS the same as its average velocity.

Motion graphs - Position vs Time (“DT graphs”)

Ball Rolling down hallway and rolling down ramp labs:

Hallway Ball Analysis WB 9-10.pdf Sample answers not our data.pdf

P5 Hallway Slow Rolling Ball DT graph P5 Hallway Fast Rolling Ball DT graph

Scatter Graphs: Line Of Best Fit

Graphing Position vs Time | Introduction to Kinematics | Physics (6:27)

Physics classroom video on: **Straight-line or constant velocity DT graphs:**
Position-Time Graphs: Constant Speed Motion (19:04)

Analyzing DT graphs:

Learning to read position-time graphs:

WB p11-13:DT graphs intro WB 9-11.pdf

3b Position-time graph#2 - solutions .pdf 3c Position-time graph#3 solutions.pdf

Extra Practice:
Read textbook pg 9
Read text 1.2 pgs 14-17 (Do Concept check pg 15, Practice probs pg 16)

Review quizlet: https://quizlet.com/ca/671874923/analyzing-dt-graphs-flash-cards/

4b Position-time graphs.pdf

Next formative quiz is on:

position, distance, displacement, speed, velocity, analyzing DT graphs without curves

DT Graphs (summary from “ball rolling” labs)

- Slope on DT graph is velocity (rise/run = displacement/time)

  Position-Time Graphs: Determining the Slope of the Line

- Straight line (unchanging slope) means constant velocity
- Curved line (changing slope) means changing velocity
- To find **average velocity**, draw a straight line between two points on the DT graph and calculate its slope
- To find **instantaneous velocity**, draw a tangent line to the curve and calculate its slope *How do you draw a tangent line* (video)

  What quantities can you find on a DT graph? Position, time, distance, displacement, speed, velocity (average and instantaneous)

Practice Analyzing Curved DT graphs [1.2]

Curved DT graphs, video: Position-Time Graphs: Changing Speed Motion

E.g. ball rolling down a ramp.Ball Rolling Down a Ramp - Lab

Answers to worksheets:

4a Finding velocity curved DT graph.pdf

4c curved DT graph solns.pdf 4 answers to curved DT graphs WBpg9.pdf

For Extras on Curved DT graphs, see sample problem/solution pg 55-57, pg 58 concept organizer, and pg 60 questions #1-3,5,6. here:

McGrawHill textbook pg 55-60 Curved DT graphs.pdf

VT Graphs (“velocity-time”)

- Show an object’s instantaneous velocity as it changes over time
- Can be generated from finding the slopes of a DT graph
- Slope on VT graph is acceleration (changing velocity over time):

  Velocity-Time Graphs: Determining the Slope (and Acceleration)

- Straight line VT graph means constant acceleration

Acceleration is how fast an object is speeding up or slowing down.

Acceleration is how fast velocity is changing over time.

Learning how to read a VT Graph [2.1]

4d understanding VT graphs note completed.pdf (WB p10)

Area under the curve of a VT graph is displacement (to learn this later)

Slope of a VT graph is acceleration

The direction of acceleration only matches the direction of motion when speeding up:

5b direction of acceleration note.pdf (WB p31 note)

WB p10-11 6 Understanding Velocity vs Time Graphs (Intro) & remote control helicopter Solutions.docx

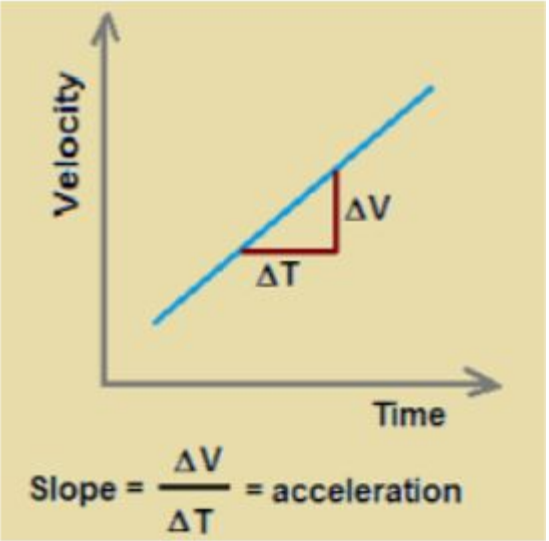
WB pg 12 VT graph T F statements hwk answers.pdf

WB p30 VT graph answers WB pg30.pdf

Extra Reading DT&VT graphs.pdf

Comparing DT and VT graphs: WB 26DT vs VT graph chart Red Car: Velocity vs. Time.pdf

What is acceleration? [text 2.1]



Watch this video: EUREKA - Acceleration part 2

Acceleration is the *change in velocity over time*.

Change in Velocity can mean:

- Speeding up OR Slowing down,

- Changing direction, even if speed is constant

Extra problems on last pg of: WB33 5 concept of acceleration note WB 16-18.pdf

Do WB p36 word problems 5b Acceleration Problems SOLUTIONS.pdf

Transition Graphs [text 1.2 (p21-24), 2.2]

From VT graph to DT graph: analyze area under the curve

-the area under each interval is the displacement or the changes in position that happen during that interval (Use this to find the next point on DT graph)

-if initial position is unknown, label the axis Displacement and start at 0

–if the interval is uniform motion, connect the points with a straight line

If the interval is constant acceleration, connect the points with a curve

From DT to VT graph: analyze slope

From VT graph to AT graph: analyze slope

Transition Graphs [text 2.2]

Lesson example: 6a transition graph lesson example.pdf

The lesson explained in video 1 & 2 below:

Finding points on DT graph by using *Areas under the Curve* (displacement) from a VT graph: 6a video 1 VT Area under Curve is displacement on DT graph.webm

Drawing the *Shape of the DT graph* by connecting the points with either a straight line or a curve: 6a video 2 Joining position points on DT graph using VT motion.webm

Solutions to homework: 6b Transition_Graphs(4)Solutions2022.pdf

Velocity to Distance (position) Time Graph Tutorial (video)

https://ophysics.com/k4b.html (simulation)

Solving motion problems using Kinematic Equations [2.3]

Two types of motions to consider:

uniform motion (= constant velocity) vs. constant acceleration



motion described by the dots on a spark timer tape (video)

The “Big 5” equations are derived from VT graphs for constant acceleration:

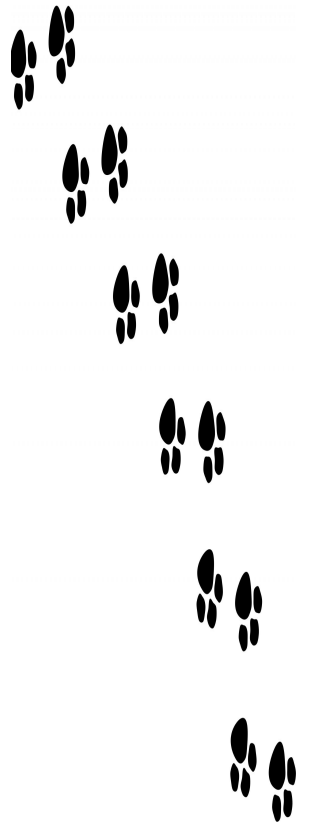
7 Deriving Kinematic Equations - completed.pdf

Problem Solving Strategies:

How to choose and use the Big 5 equations: **”It’s as easy as 3-4-5!”**

Picture the motion

Kinematics Equations Examples - complete2024.pdf



How to draw a Well-Labelled Line Diagram

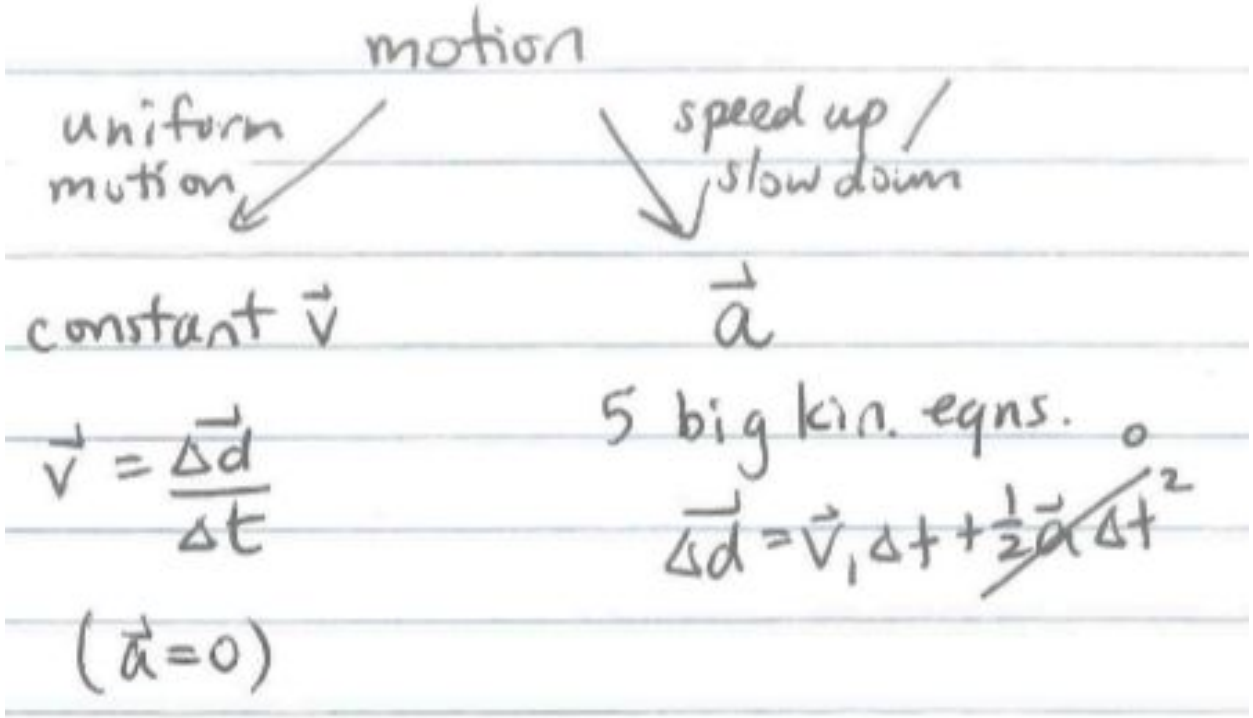
Helps you picture the motion to solve the problem. Make it big enough to fit your labels and given quantities. You want to label all givens and important information (include the unknowns you need).

Start drawing an arrow to represent the motion. Label the displacement, time interval, and any acceleration happening during this interval.

Number the positions at the beginning and end of the arrow. Instantaneous velocities would be labelled at these positions.

As you start filling in the diagram with vector information, you need to include a sign/direction convention, e.g. “Let down be positive.” Then you can enter any vector quantities as “+/-” as you label them.

For each interval you analyze, decide whether the motion is **uniform** or **accelerated**:



Types of Kinematic Problems

Vertical Motion in Free Fall: [text 3.1]

-objects in “Free Fall” accelerate at a constant **acceleration due to gravity** of 9.81 m/s/s on Earth.

The Acceleration of Gravity 5.1 Dealing With Gravity blank note.docx

Consecutive intervals:

-an object's consecutive displacements show different types of motion, that need to be analyzed separately

-consecutive intervals have overlapping information

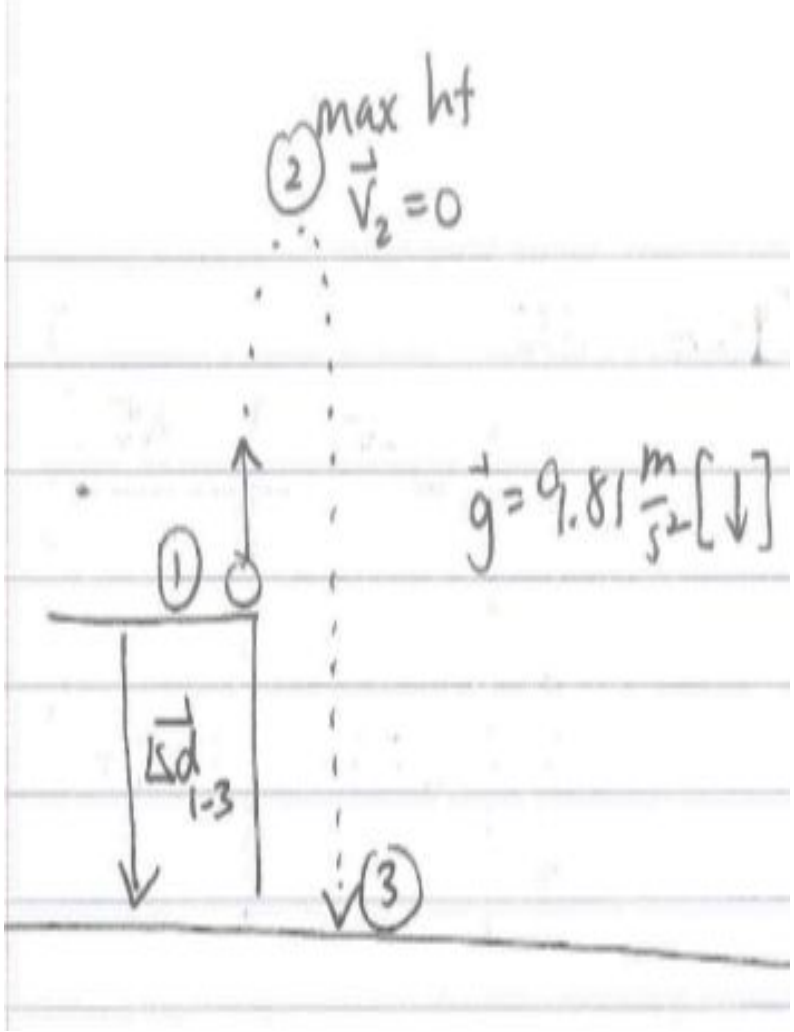
Chase & Collision:

-two objects move in the same direction and one catches up to the other, at the same place and time

-two objects move towards each other and collide/meet, at the same place and time

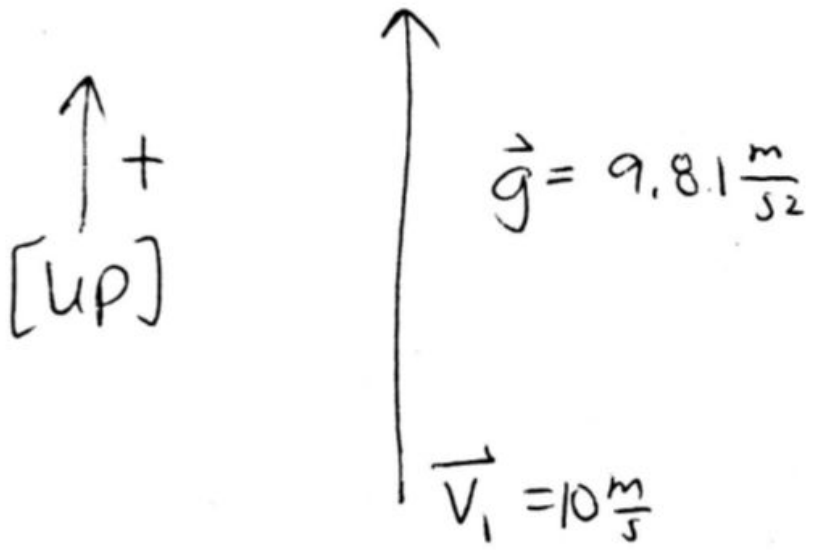
Vertical Motion

Unless stated otherwise, assume little to no air resistance. This means objects moving in the air are "Free-Falling" and they accelerate due to gravity.



Formative quiz - Find and describe errors in the work below:

Question: A ball is thrown up at 10 m/s. Find the time it takes to reach a height of 3 m.



$$\vec{a} = \frac{\vec{\Delta v}}{\Delta t}$$

$$\Delta t = \frac{\vec{\Delta v}}{\vec{a}} = \frac{10 \frac{m}{s}}{9.81 \frac{m}{s^2}} = \ldots$$

As learning

Acceleration due to Gravity Lab

The meaning of “free fall”

Minds on: How can we prove that an object is free-falling?

(How would you design a lab to prove that an object is free-falling?)



Full Solutions Checklist

Diagram has sign convention for direction

Vector symbols have arrows

State chosen equation,

then rearranged equation if needed

Substitute values with units (sometimes a unit conversion is needed)

State unrounded answer, then final answer rounded to correct number of Sig Figs

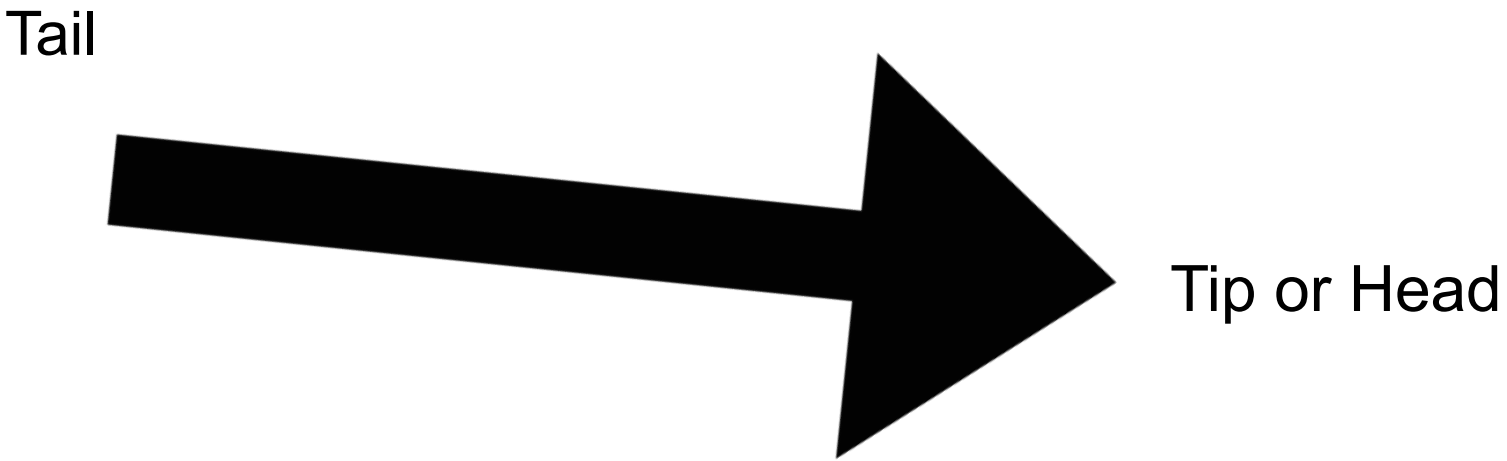
Vector answers have units and [direction]

Only round ONCE per final answer!

Graphs Checklist

Title of graph
Labels for axes
Even scales
Show data points clearly
Draw curve or line of best fit correctly
Draw tangent lines long enough and show the point of contact on the curve
Circle the 2 points used in any slope calculation

Anatomy of a Vector: Label the tip and the tail.

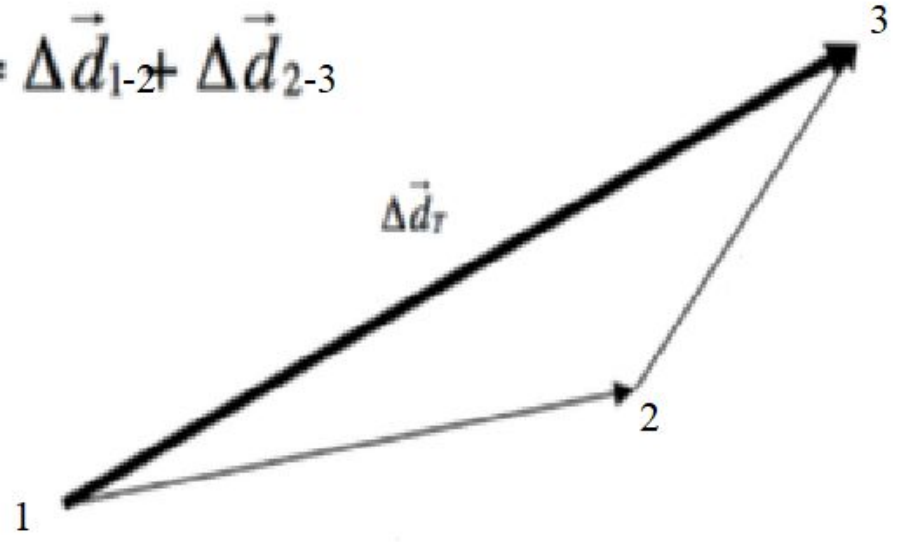


Have we added vectors before?

A = YES B = NO

Yes, we have added displacement vectors in series to get a total displacement.

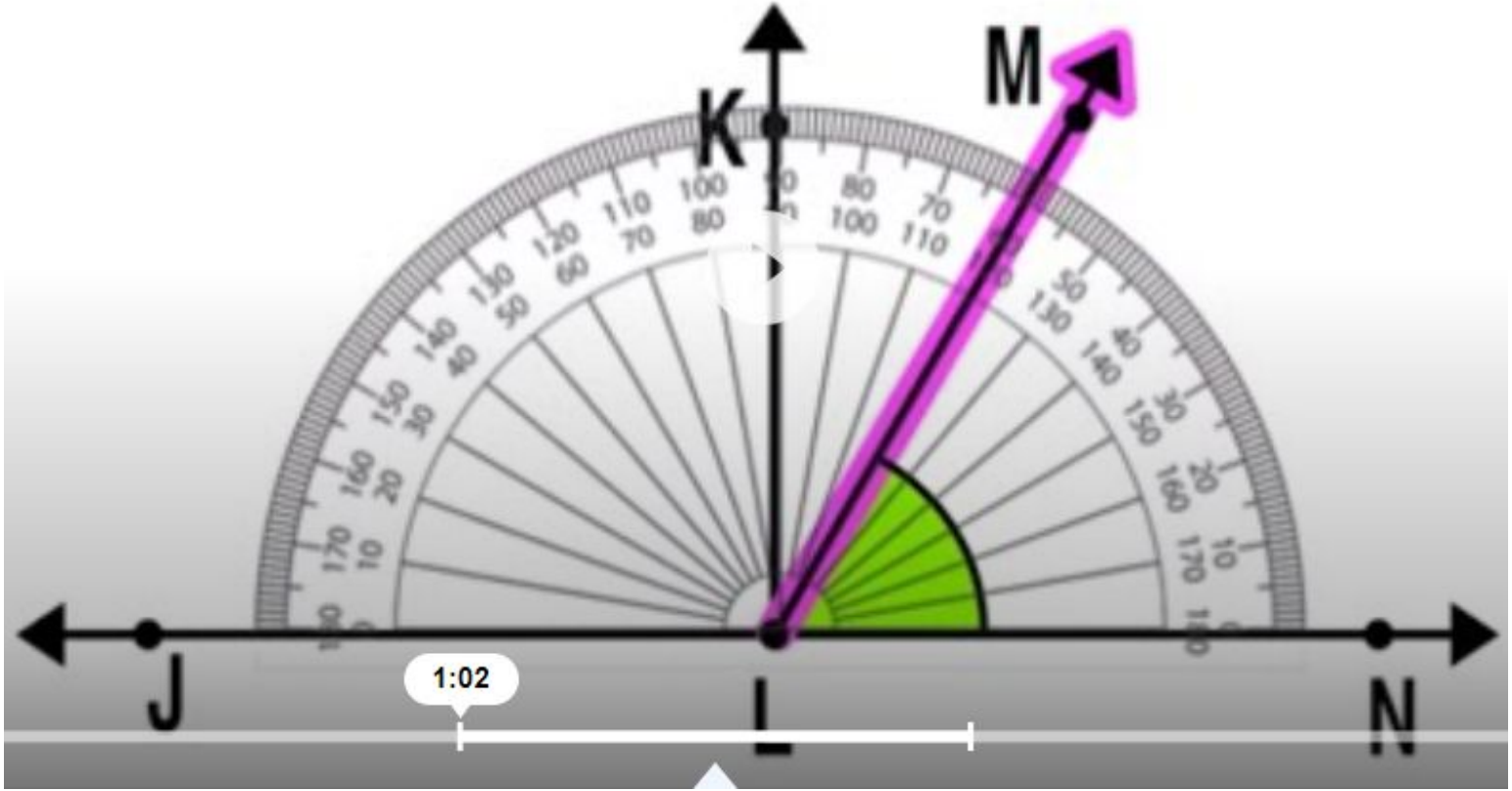
$$\Delta\vec{d}_T = \Delta\vec{d}_{1\text{-}2} + \Delta\vec{d}_{2\text{-}3}$$



We add vectors
TIP to TAIL.

Watch video with cc:

Do you remember how to use a protractor?



Adding and Subtracting Vectors Graphically

Adding Vectors Graphically This video shows:

- how to add vectors graphically by connecting them tip/head to tail

- that vectors can be added in any order; the result will be the same

- how to subtract vectors by “adding the opposite” first

Graphical Method of Vector Addition This video shows how you add vectors graphically. It shows a full solution that you can use for practice.

Analytical Method of Vector Addition Can start watching video from 6:06. This video show you how to add vectors by component method. It summarizes and shows a full solution that you can use for practice.

Vector Lesson Notes/Handout

8a vectors 2D lesson 1.pdf 8b Vectors 2D addition - answers

Vector Calculator

Solving problems using scale diagrams

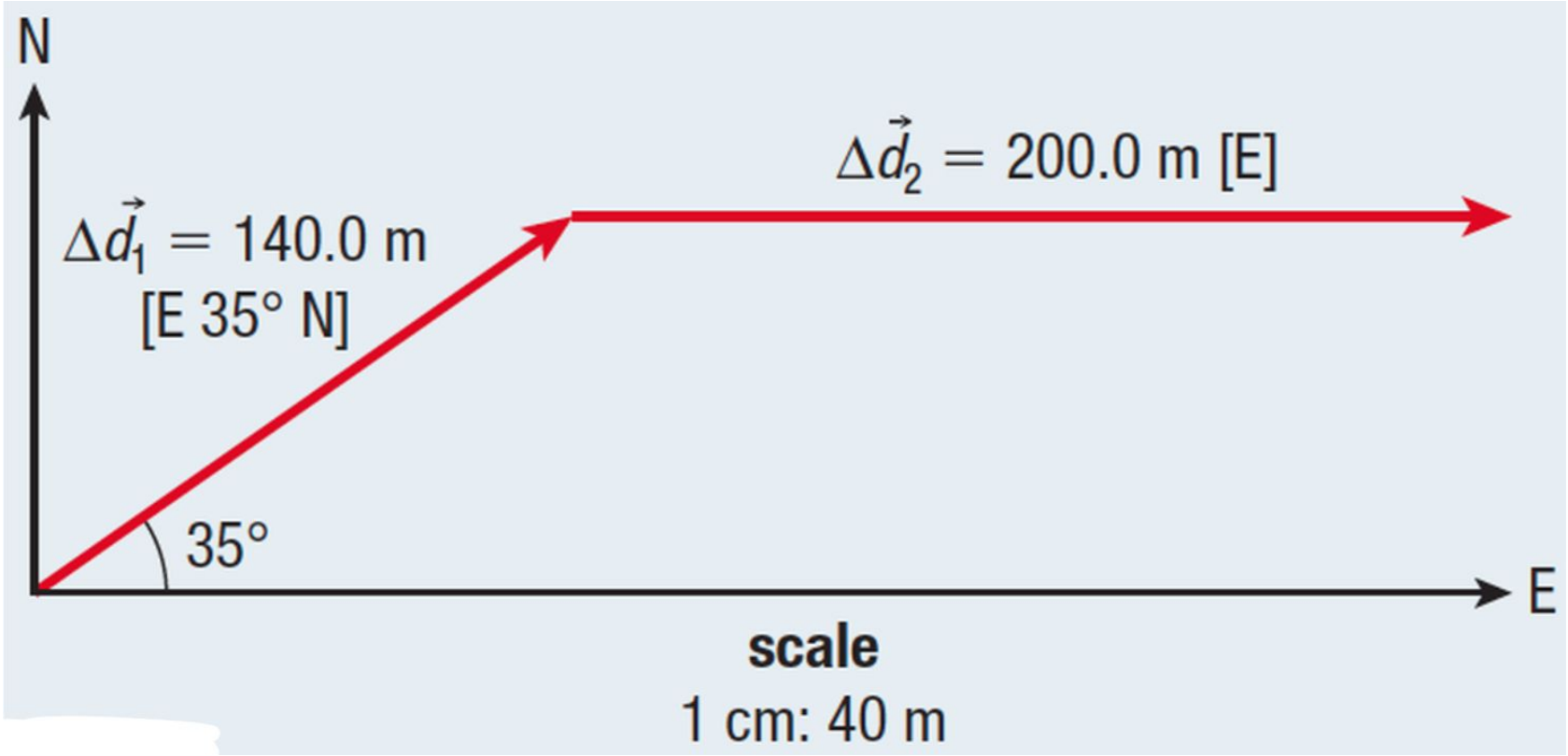
1. Write out the full vector equation.
2. Include the scale that is being used (eg. 1 cm : 10 km).
3. Draw the diagram using a ruler and protractor.
4. State the resultant using proper units and direction.

Sample question:

Suppose you walk to a friend's house, taking a shortcut across a field. Your first displacement is 140 m [E 35° N] across the field. Then you walk 200.0 m [E] along the sidewalk. Determine your total displacement using a scale diagram.

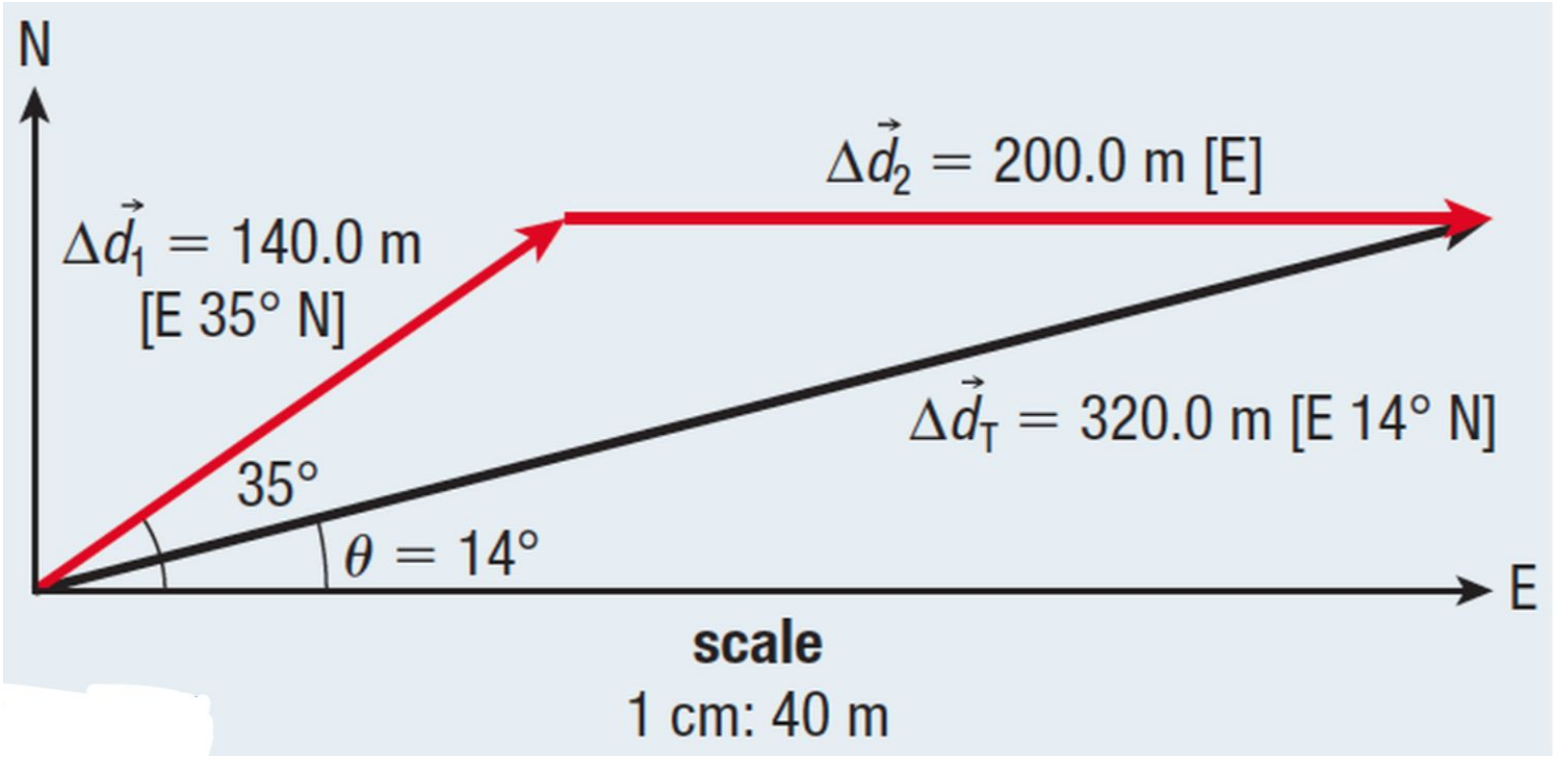
More practice: 8c_Vector_Problems_2022.pdf

Find the total displacement by drawing a scale diagram:



Add the series of displacements by connecting them tip to tail.

The total displacement is from the initial position (tail of 1st vector) to final position (head of last vector).



The angle at the ***tail*** (not the head) determines the direction of the vector.

There many correct ways to describe a vector's direction:

[14° N of E]

↑ =

$\Delta\vec{d}_T = 320$ m [E 14° N]

OR

[N 76° E] = [76° E of N]

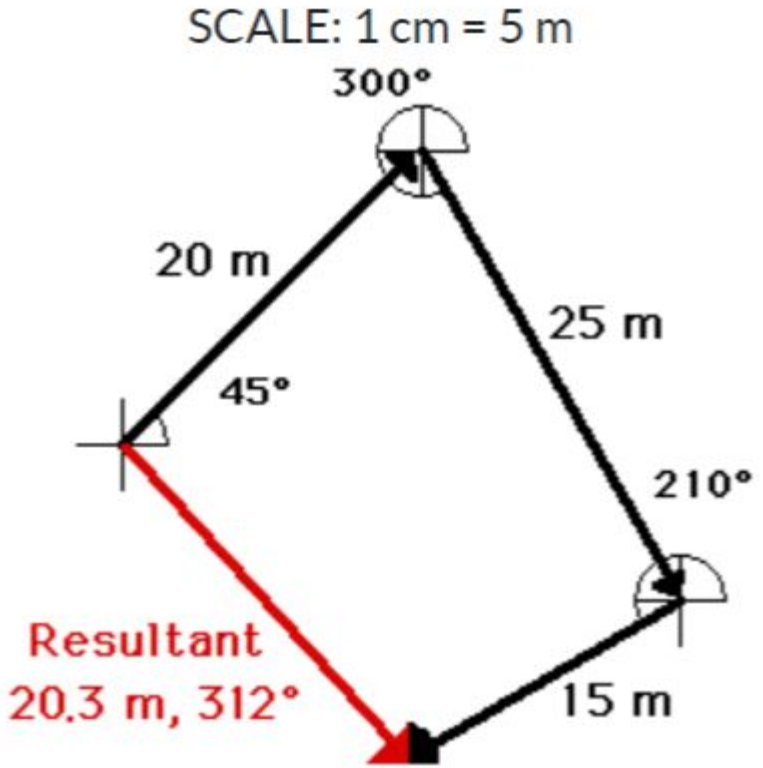
Practice

See if you can add these 3 displacement vectors algebraically and compare your answer to the resultant shown.

20 m [NE]

25 m [E 60 deg S]

15 m [W 30 deg S]



Does order matter, when adding vectors?

A = YES B = NO

Vector Components This video summarizes:

- how we define the x and y components of a vector using trig ratios

- how we use Pythagorean theorem to find the magnitude of a vector, given the magnitudes of its x and y components

- how we use a trig ratio (tan) to find the angle of a vector, given the magnitudes of its x and y components {This angle would then be used to describe the direction of our vector}

Unit 1 Test Topics

SDs/Rounding/Rearranging formulas/Unit conversion (intro skills that show up throughout the entire course)

Position, displacement, distance

Speed, velocity (average, instantaneous), acceleration

Analyzing Motion graphs: DT graph, VT graph, AT graph

Transition graphs (given VT graph, generate DT and AT graph)

Use Kinematic Equations to Solve motion problems (vertical motion, consecutive intervals, chase & collision)

<u>Day 2:</u>

2D Vectors Quiz: Add and subtract vectors using scale diagram (graphical method) and using vector components (algebraic method)

Gravity lab quiz: based on lab skills/analysis skills of your Gravity lab